

US Domestic Flight Performance Analysis

Data Analytics Capstone Project Report

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Tools Used: Python, PostgreSQL, Power BI Desktop

1. Executive Summary

This project analyses U.S. domestic flight performance data from 2015 (approximately 5.8 million records). The objective was to clean the data, load it into PostgreSQL, build analytical views, and create an interactive Power BI dashboard to uncover patterns in delays, cancellations, airline performance, airport operations, and temporal trends.

Key Highlights:

- Weather is the leading cause of cancellations (54.35%).
 - Hawaiian Airlines performed the best in on-time performance.
 - Spirit Airlines had the worst on-time performance.
 - Evening flights generally show lower OTP rates.
 - Atlanta airport handles the highest volume of flights.
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2. Introduction

Project Objective To build an end-to-end data analytics solution that helps understand flight delays and cancellations across U.S. airlines and airports.

Business Questions Addressed

- Which airlines perform best and worst in terms of on-time performance?
 - What are the main causes of flight cancellations and delays?
 - How do performance metrics vary by month, day, and time of day?
 - Which airports and routes experience the highest delays?
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3. Data Source

- **Dataset:** U.S. Flight Delays (2015)

- **Source:** Kaggle (originally from U.S. Department of Transportation)
 - **Size:** ~5.8 million rows
 - **Tables:** flights, airlines, airports
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4. Tools & Technologies Used

Tool	Purpose
Python	Data cleaning & loading
PostgreSQL	Data storage, transformation & views
Power BI	Interactive dashboard & visualization
DAX	Custom measures in Power BI

5. Data Cleaning & Transformation (Python + SQL)

5.1 Python Cleaning Steps

- Handled missing values in delay columns (filled with 0)
- Converted time columns (HHMM) into minutes
- Created CANCELLATION_REASON_DESC
- Created FLIGHT_DATE

5.2 SQL Transformation Steps

- Added derived columns (airline_name, origin_airport_name, etc.)
- Created timestamp columns
- Built analytical view: flight_analysis

Space for SQL Query Outputs:

Figure 5.1: SQL Query – Overall Flight Performance Summary



Figure 5.2: 2. Airline Performance Analysis

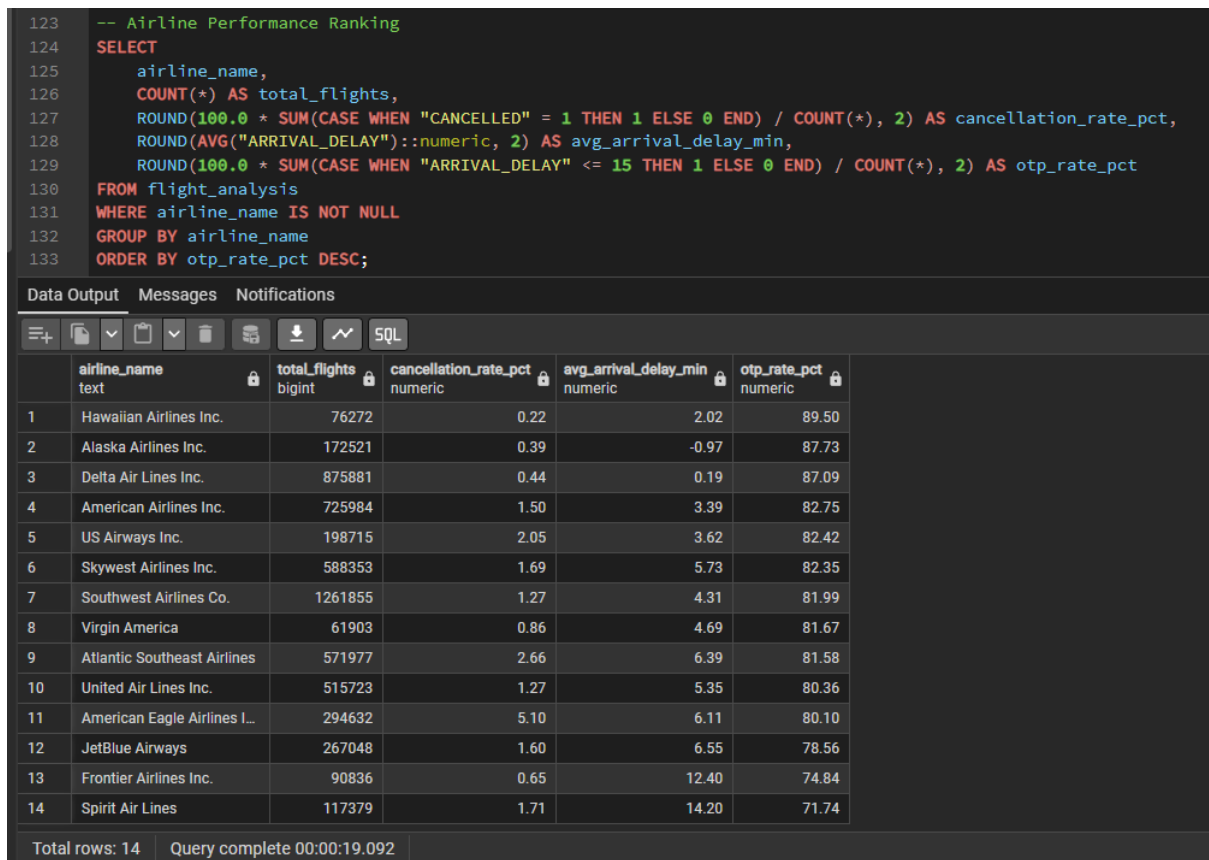


Figure 5.3: Monthly Performance Trends

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135 -- Monthly Performance Analysis
136 SELECT
137     "MONTH",
138     COUNT(*) AS total_flights,
139     ROUND(AVG("ARRIVAL_DELAY")::numeric, 2) AS avg_arrival_delay_min,
140     ROUND(100.0 * SUM(CASE WHEN "CANCELLED" = 1 THEN 1 ELSE 0 END) / COUNT(*), 2) AS cancellation_rate_pct,
141     ROUND(100.0 * SUM(CASE WHEN "ARRIVAL_DELAY" <= 15 THEN 1 ELSE 0 END) / COUNT(*), 2) AS otp_rate_pct
142 FROM flight_analysis
143 GROUP BY "MONTH"
144 ORDER BY "MONTH";

```

	MONTH integer	total_flights bigint	avg_arrival_delay_min numeric	cancellation_rate_pct numeric	otp_rate_pct numeric
1	1	469968	5.65	2.55	80.34
2	2	429191	7.90	4.78	78.61
3	3	504312	4.80	2.18	81.81
4	4	485151	3.12	0.93	83.76
5	5	496993	4.42	1.15	82.62
6	6	503897	9.39	1.81	77.77
7	7	520718	6.35	0.92	80.03
8	8	510536	4.55	0.99	82.22
9	9	464946	-0.77	0.45	87.63
10	10	486165	-0.78	0.50	88.23
11	11	467972	1.09	0.98	85.56
12	12	479230	5.97	1.68	80.53

Figure 5.4: Day of Week Analysis

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146 -- Performance by Day of Week
147 SELECT
148     "DAY_OF_WEEK",
149     COUNT(*) AS total_flights,
150     ROUND(AVG("ARRIVAL_DELAY")::numeric, 2) AS avg_arrival_delay_min,
151     ROUND(100.0 * SUM(CASE WHEN "CANCELLED" = 1 THEN 1 ELSE 0 END) / COUNT(*), 2) AS cancellation_rate_pct,
152     ROUND(100.0 * SUM(CASE WHEN "ARRIVAL_DELAY" <= 15 THEN 1 ELSE 0 END) / COUNT(*), 2) AS otp_rate_pct
153 FROM flight_analysis
154 GROUP BY "DAY_OF_WEEK"
155 ORDER BY "DAY_OF_WEEK";

```

	DAY_OF_WEEK bigint	total_flights bigint	avg_arrival_delay_min numeric	cancellation_rate_pct numeric	otp_rate_pct numeric
1	1	865543	5.84	2.43	81.45
2	2	844600	4.16	1.78	82.74
3	3	855897	3.79	1.25	82.76
4	4	872521	5.57	1.41	81.07
5	5	862209	4.70	1.02	81.80
6	6	700545	1.82	1.25	84.86
7	7	817764	3.88	1.61	82.70

Figure 5.5: Time of Day Analysis

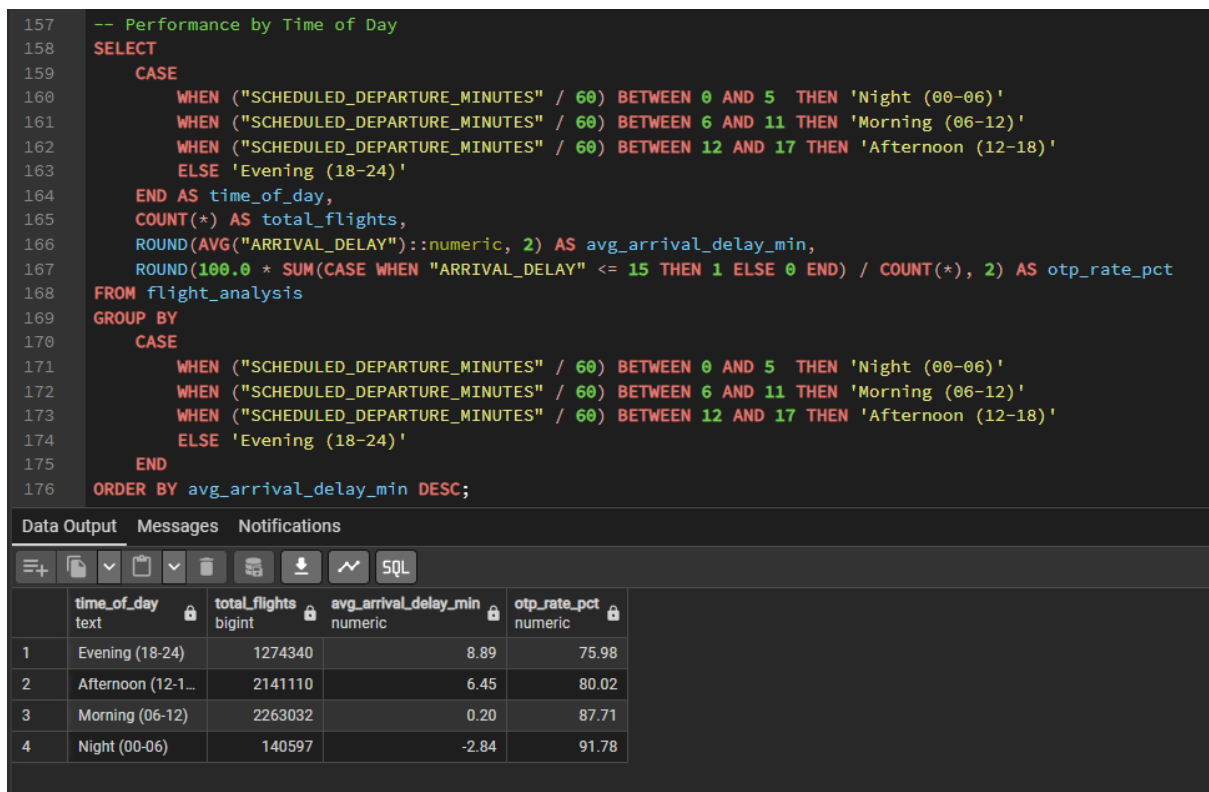


Figure 5.6: Cancellation Analysis

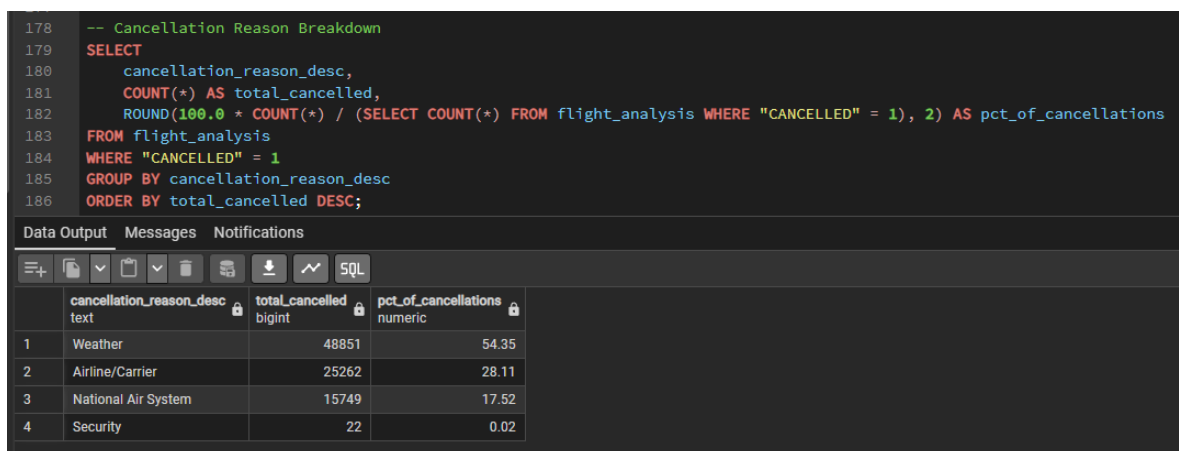


Figure 5.7: Delay Type Contribution

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188 -- Delay Type Contribution (Total Minutes)
189 SELECT 'Airline Delay' AS delay_type, SUM("AIRLINE_DELAY") AS total_minutes FROM flight_analysis
190 UNION ALL
191 SELECT 'Weather Delay', SUM("WEATHER_DELAY") FROM flight_analysis
192 UNION ALL
193 SELECT 'NAS Delay', SUM("AIR_SYSTEM_DELAY") FROM flight_analysis
194 UNION ALL
195 SELECT 'Security Delay', SUM("SECURITY_DELAY") FROM flight_analysis
196 UNION ALL
197 SELECT 'Late Aircraft Delay', SUM("LATE_AIRCRAFT_DELAY") FROM flight_analysis
198 ORDER BY total_minutes DESC;
```

Data Output

Messages

Notifications

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SQL

	delay_type text	total_minutes double precision
1	Late Aircraft Del...	24961931
2	Airline Delay	20172956
3	NAS Delay	14335762
4	Weather Delay	3100233
5	Security Delay	80985

Figure 5.8: Airport Performance

200 -- Top 15 Busiest Airports

201 SELECT

202 origin_airport_name,

203 origin_city,

204 origin_state,

205 COUNT(*) AS total_flights,

206 ROUND(AVG("ARRIVAL_DELAY")::numeric, 2) AS avg_arrival_delay_min,

207 ROUND(100.0 * SUM(CASE WHEN "CANCELLED" = 1 THEN 1 ELSE 0 END) / COUNT(*), 2) AS cancellation_rate_pct

208 FROM flight_analysis

209 GROUP BY origin_airport_name, origin_city, origin_state

210 ORDER BY total_flights DESC

211 LIMIT 15;

Data Output

Messages

Notifications

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Figure 5.9: Route Analysis

213	-- Top 15 Worst Delay Routes (Minimum 100 flights)
214	SELECT
215	origin_airport_name ' → ' dest_airport_name AS route,
216	COUNT(*) AS total_flights,
217	ROUND(AVG("ARRIVAL_DELAY")::numeric, 2) AS avg_arrival_delay_min
218	FROM flight_analysis
219	GROUP BY origin_airport_name, dest_airport_name
220	HAVING COUNT(*) >= 100
221	ORDER BY avg_arrival_delay_min DESC
222	LIMIT 15;
223	

Data Output	Messages	Notifications
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	route text	total_flights bigint	avg_arrival_delay_min numeric
1	John F. Kennedy International Airport (New York International Airport) → Eagle County Regional Airp...	110	36.28
2	Oakland International Airport → George Bush Intercontinental Airport	179	33.08
3	Aspen-Pitkin County Airport → Dallas/Fort Worth International Airport	335	31.26
4	Atlantic City International Airport → Detroit Metropolitan Airport	147	30.48
5	Sarasota-Bradenton International Airport → LaGuardia Airport (Marine Air Terminal)	433	29.26
6	Atlantic City International Airport → Gen. Edward Lawrence Logan International Airport	182	29.10
7	Minneapolis-Saint Paul International Airport → Trenton Mercer Airport	157	29.07
8	Dallas/Fort Worth International Airport → Kahului Airport	504	28.34
9	Dallas/Fort Worth International Airport → Honolulu International Airport	697	28.20
10	Yeager Airport → Dallas/Fort Worth International Airport	146	27.99
11	Eagle County Regional Airport → Chicago O'Hare International Airport	109	27.07
12	Pittsburgh International Airport → John F. Kennedy International Airport (New York International Air...	164	26.84
13	Myrtle Beach International Airport → Philadelphia International Airport	194	26.17
14	Orlando International Airport → Trenton Mercer Airport	486	26.14
15	Chicago O'Hare International Airport → Honolulu International Airport	334	26.06

Total rows: 15	Query complete 00:00:18.194
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Figure 5.10: Bonus: Monthly Delay Trend

224	-- Monthly Trend of Average Delay
225	SELECT
226	"MONTH",
227	ROUND(AVG("ARRIVAL_DELAY")::numeric, 2) AS avg_arrival_delay,
228	ROUND(AVG("DEPARTURE_DELAY")::numeric, 2) AS avg_departure_delay
229	FROM flight_analysis
230	GROUP BY "MONTH"
231	ORDER BY "MONTH";

Data Output	Messages	Notifications
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	MONTH integer	avg_arrival_delay numeric	avg_departure_delay numeric
1	1	5.65	9.52
2	2	7.90	11.33
3	3	4.80	9.46
4	4	3.12	7.65
5	5	4.42	9.35
6	6	9.39	13.74
7	7	6.35	11.30
8	8	4.55	9.84
9	9	-0.77	4.80
10	10	-0.78	4.96
11	11	1.09	6.88
12	12	5.97	11.59

Total rows: 12	Query complete 00:00:17.181
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6. Database Design & Analytical View

Created View: flight_analysis This view joins flights, airlines, and airports tables and includes denormalized columns for better Power BI performance.

7. Power BI Dashboard Development

The dashboard contains **6 pages** with consistent dark theme design.

Page 1: Home / Executive Overview

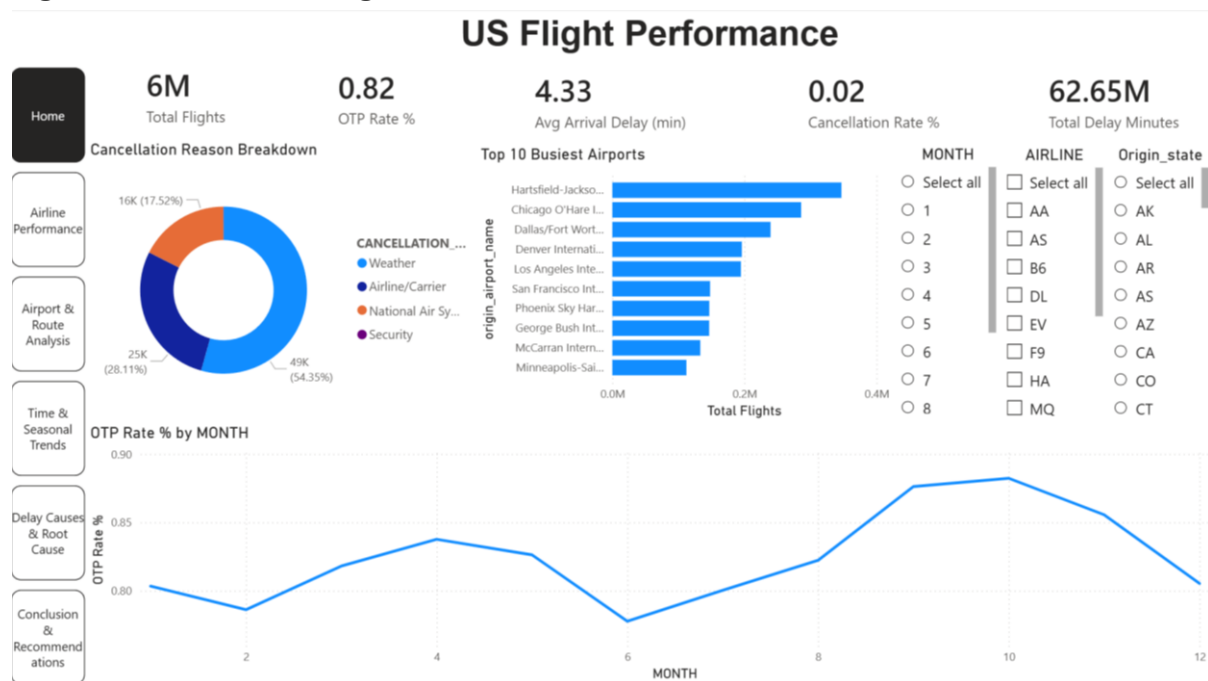
Purpose: High-level snapshot of overall performance.

Key Visuals:

- 5 KPI Cards
- Cancellation Reason Donut
- Top 10 Busiest Airports
- Monthly OTP Trend

Space for Screenshot:

Figure 7.1: Power BI – Page 1: Home Dashboard



Page 2: Airline Performance

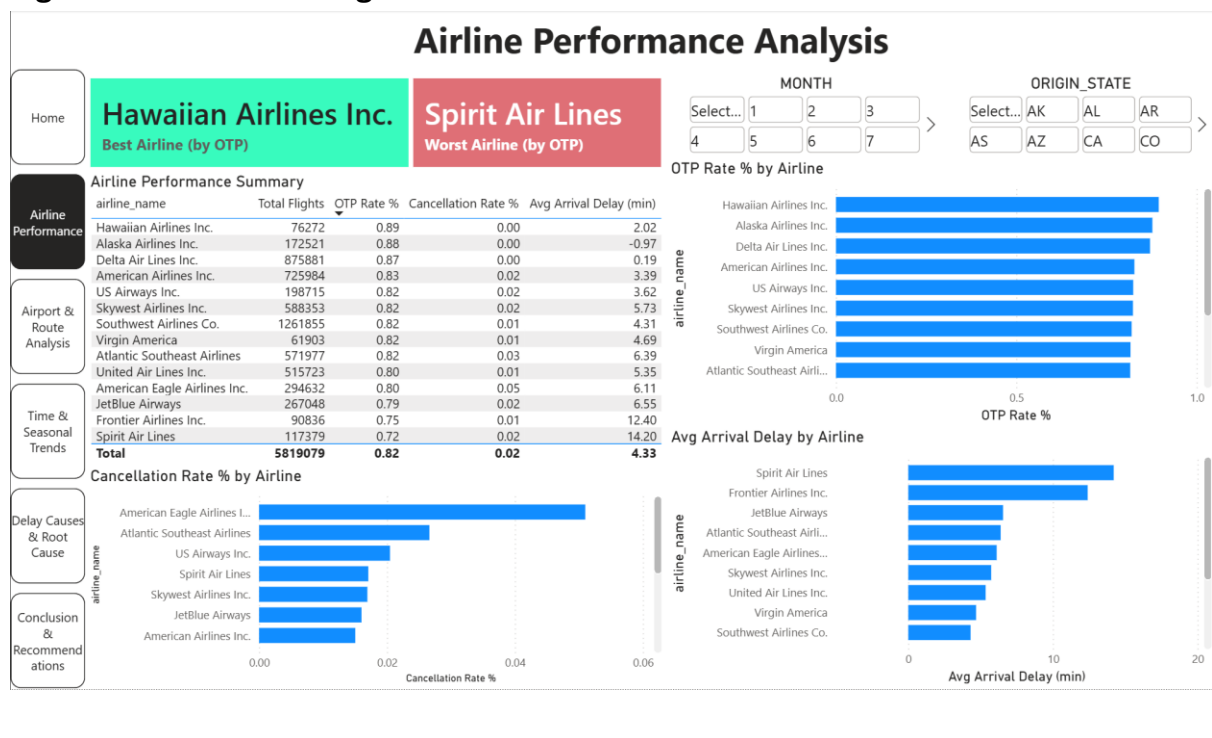
Purpose: Compare airlines on OTP, delays, and cancellations.

Key Visuals:

- Best & Worst Airline Cards
- Airline Performance Summary Table
- OTP Rate % by Airline
- Cancellation Rate by Airline

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Figure 7.2: Power BI – Page 2: Airline Performance



Page 3: Airport & Route Analysis

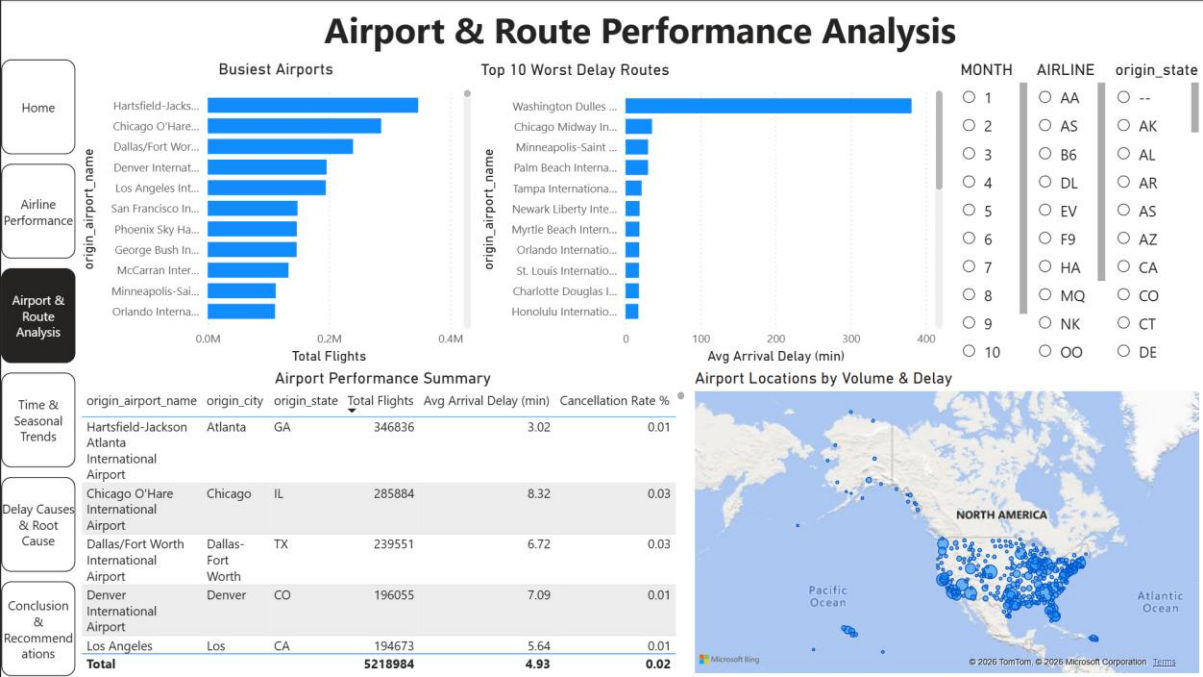
Purpose: Analyze airport-level performance and route delays.

Key Visuals:

- 3 KPI Cards (Busiest Airport, Highest Delay Airport, Busiest Route)
- Top 15 Busiest Airports
- Airport Performance Summary Table
- Map: Airport Locations by Volume & Delay
- Top 10 Worst Delay Routes

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Figure 7.3: Power BI – Page 3: Airport & Route Analysis



Page 4: Time & Seasonal Trends

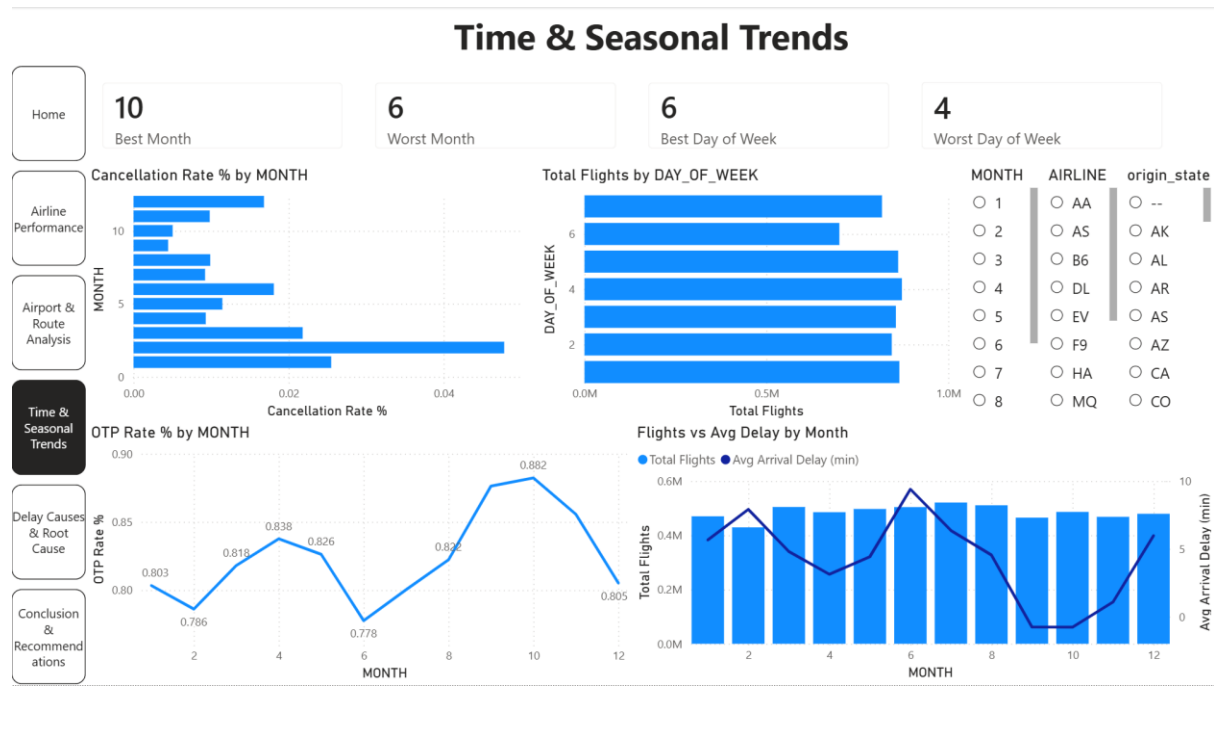
Purpose: Understand performance patterns over time.

Key Visuals:

- Best & Worst Month/Day Cards
- OTP Rate % by Month
- Cancellation Rate by Month
- Flights by Day of Week
- OTP Rate by Time of Day

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Figure 7.4: Power BI – Page 4: Time & Seasonal Trends



Page 5: Delay Causes & Root Cause

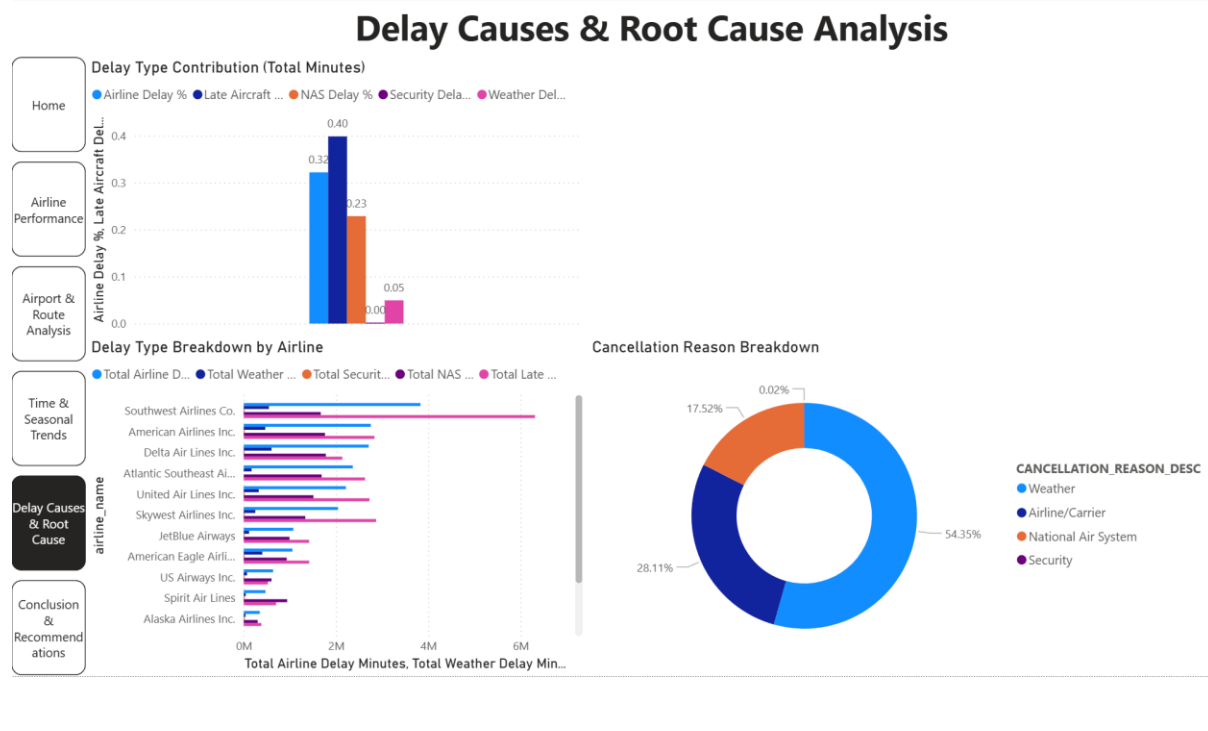
Purpose: Deep dive into reasons behind delays and cancellations.

Key Visuals:

- Cancellation Reason Breakdown (Donut)
- Delay Type Contribution (Donut)
- Delay Contribution % by Type
- Delay Type Breakdown by Airline

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Figure 7.5: Power BI – Page 5: Delay Causes & Root Cause



Page 6: Conclusion & Recommendations

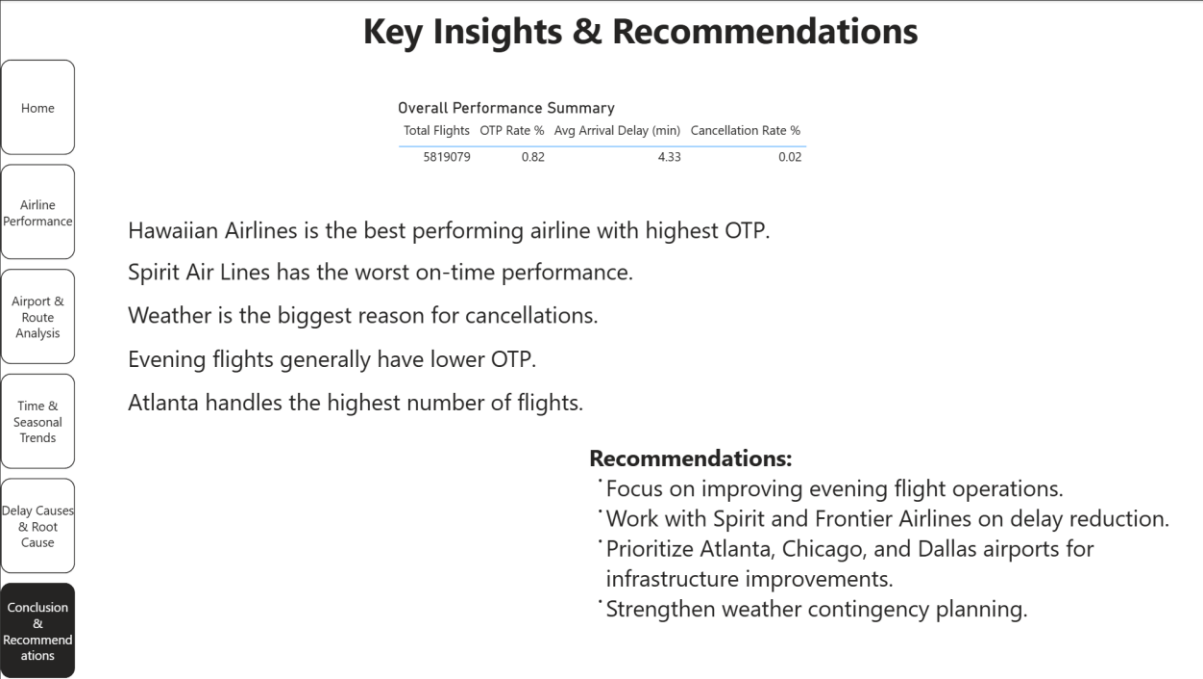
Purpose: Summarize key findings and provide actionable recommendations.

Key Visuals:

- 6 Summary KPI Cards
- Overall Performance Summary Table
- Key Highlights
- Recommendations Text Box

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Figure 7.6: Power BI – Page 6: Conclusion & Recommendations



8. Key Findings & Insights

Major Insights:

- Weather causes the highest number of cancellations.
- Hawaiian Airlines has the best on-time performance.
- Spirit Airlines has the worst on-time performance.
- Evening flights (after 6 PM) show lower OTP rates.
- Atlanta airport handles the maximum number of flights.

9. Recommendations

1. Airlines should focus on improving evening flight operations.
2. Spirit and Frontier should prioritize delay reduction strategies.
3. Airports like Atlanta, Chicago, and Dallas need better delay management systems.
4. Stronger weather contingency planning is required across airlines.

10. Conclusion

This project successfully demonstrated the complete data analytics lifecycle — from data cleaning in Python, transformation in PostgreSQL, to building an interactive and insightful dashboard in Power BI. The dashboard provides actionable insights that can help airlines and airports improve operational efficiency.

11. Limitations

- Data is limited to the year 2015 only.
 - Some airport codes were missing in the original dataset.
 - Real-time data is not available (static dataset).
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12. Future Scope

- Add predictive analytics (predict flight delays).
 - Include more years of data for trend analysis.
 - Integrate live flight data using APIs.
 - Add role-based access in Power BI Service.
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Appendix

Appendix A: [SQL Scripts](#)

Appendix B: [Python Script](#)

Appendix C: [Power BI File](#)